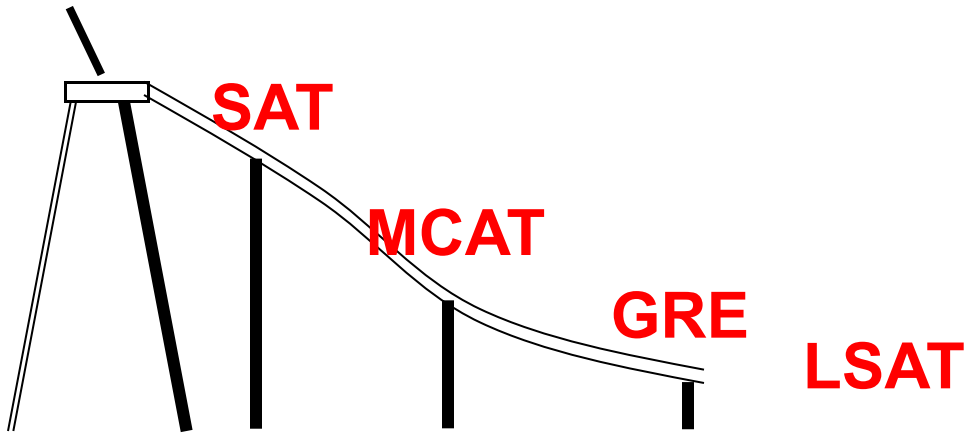
# Test Slide



# Quantum Computing

## (or how we learned to stop worrying and love quantum theory.)




**Dave Bacon**

*Department of Physics*

*Institute for Quantum Information*

California Institute of Technology

A sudden bold and unexpected question doth many times surprise a man and lay him open.

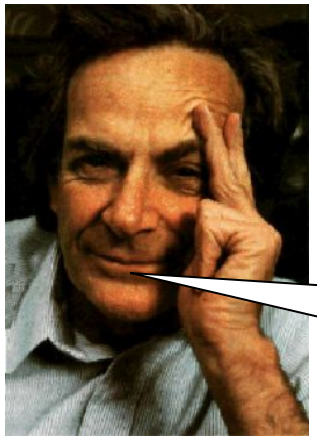
A prudent question is one-half of wisdom.



Sir Francis Bacon
(1561-1628)

?

# Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965

Anyone who is not shocked by quantum theory has not understood it.



Niels Bohr
Nobel Prize 1922



They understand it.
They use it.
But they don’t really believe it?

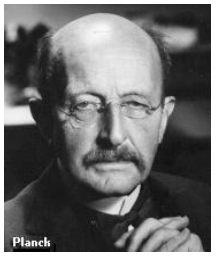
# Quantum History

Inventin

Worrying

Acceptin g

1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000




Max Planck



Niels Bohr



Erwin Schrodinger



John Von Neumann



John Bell



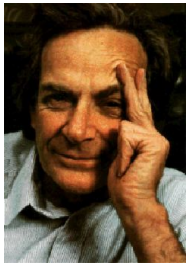
Peter Shor



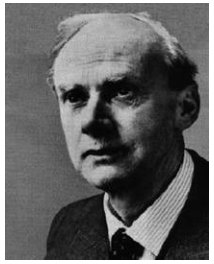
Louis de Broglie



Werner Heisenberg



Richard Feynman



P.A.M. Dirac

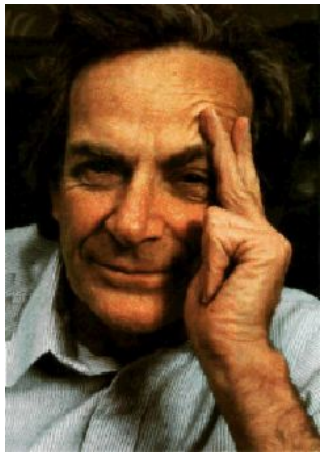
# Worrying?




"I have this student, he's thinking about the foundations of quantum mechanics. He is doomed." — **John McCarthy** (of Artificial Intelligence fame)

# Acceptance

"Do not ask yourself, if you can possibly avoid that, 'how can it be like that?' because you will lead yourself down a blind alley in which no one has ever escaped."



"For a successful technology, nature must take precedence over public relations, for nature cannot be fooled."

Richard Feynman

"How can we simulate quantum mechanics?...Can we do it with a new kind of computer…a quantum computer? It's not a Turing machine but a machine of a different kind" (1981)

"Perhaps…we need a mathematical theory of quantum automata…" – Yu Manin (Radio Moscow, 1980)

What IF we think about COMPUTERS which operate according to QUANTUM THEORY?




QUANTUM THEORY rules but its intricacies become unimportant for effects at “larger sizes.”

# Near Misses



Alan Turing

Rediscovered the Zeno effect in quantum theory (if you measure a system constantly it will not evolve!)



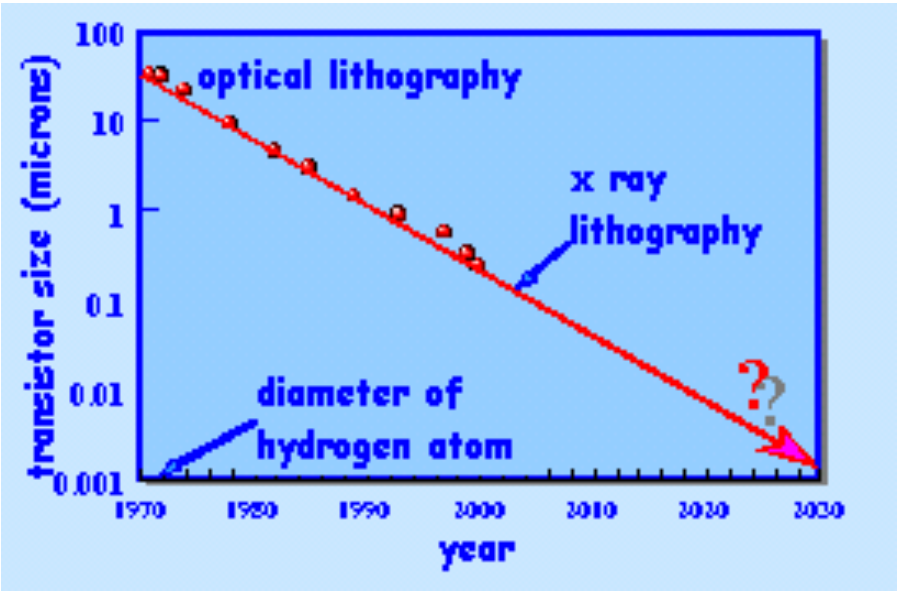
John Von Neumann

Wrote classic text on the axioms of quantum theory.

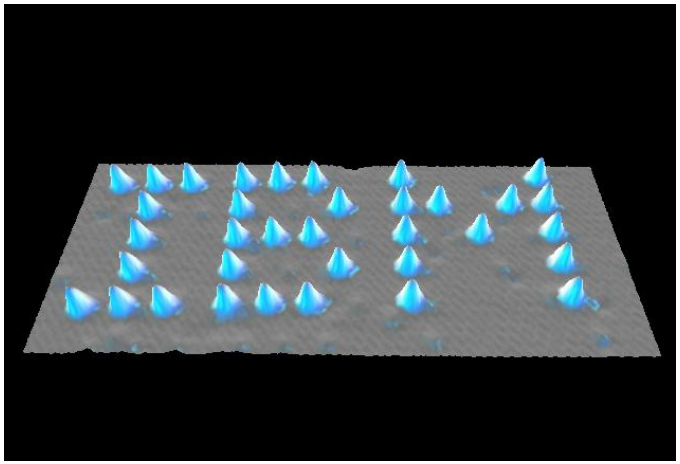
# We Needed Moore



Noyce, Grove, Moore



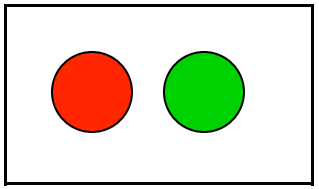

Moore's Law
(Moore's prophesy?)



# Quantum Uncertainty Is Not Special

Quantum theory gives us probabilities of outcomes.

50% of the time the red light comes on
50% of the time the green light comes on



Try to describe classical computers.

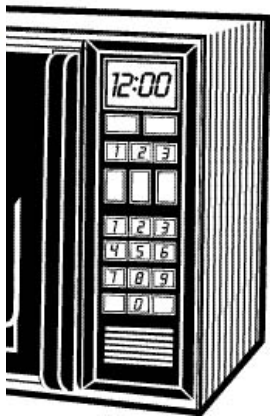
Hardware errors do occur, but the timescales involved are ridiculously long.




99.99999999999999999% the computer works perfectly
0.00000000000000001 the computer makes a mistake

# Randomizing Microwave






Digital Coffee
(Not Java!)

Mystery Markov Microwave







**Scalding Hot**

**H**

**Freezing Cold**

**C**

# Marko

**The true method of knowledge is experiment. - William Blake 1788**

**V**

- Run Experiments To Understand MMM Machine

If you put in C, 70% of the time you get H out
and 30% of the time you get C out
If you put in H, 80% of the time you get H out
and 20% of the time you get C out

- A nice little formalism

**H** $\longrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

**C** $\longrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$M_{mm} = \begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}$$

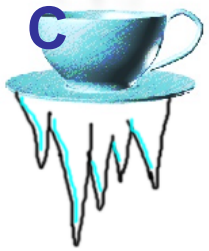
columns sum to 1

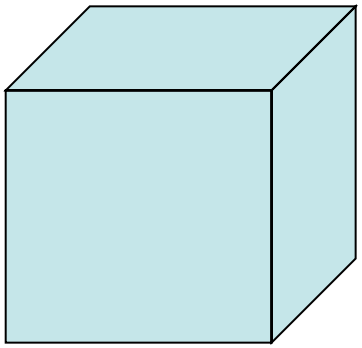
$$\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix}$$

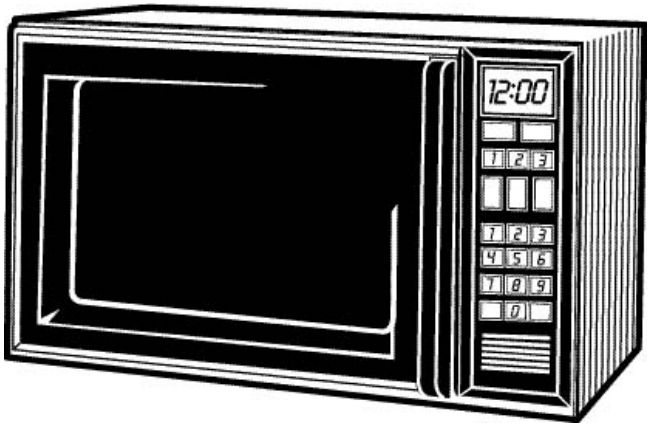
$$\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.7 \\ 0.3 \end{pmatrix}$$

# Uncertainty

20%
C



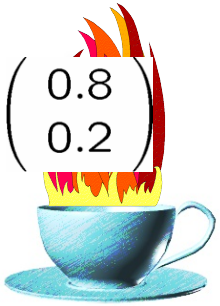





22%
C



$$0.8 \times 0.8 + 0.2 \times 0.7 = 0.78$$




$$\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0.78 \\ 0.22 \end{pmatrix}$$

$$0.8 \times 0.2 + 0.2 \times 0.3 = 0.22$$

80%
H

78%
H

# Markov Chains

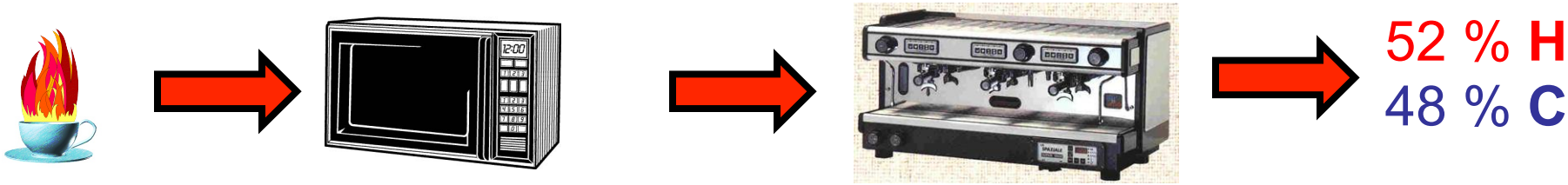



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} \quad \begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}\begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0.78 \\ 0.22 \end{pmatrix}$$

or

$$\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} = \begin{pmatrix} 0.78 & 0.77 \\ 0.22 & 0.23 \end{pmatrix} \quad \begin{pmatrix} 0.78 & 0.77 \\ 0.22 & 0.23 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.78 \\ 0.22 \end{pmatrix}$$

---




$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} \quad \begin{pmatrix} 0.5 & 0.6 \\ 0.5 & 0.4 \end{pmatrix}\begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0.52 \\ 0.48 \end{pmatrix}$$

or

$$\begin{pmatrix} 0.5 & 0.6 \\ 0.5 & 0.4 \end{pmatrix}\begin{pmatrix} 0.8 & 0.7 \\ 0.2 & 0.3 \end{pmatrix} = \begin{pmatrix} 0.52 & 0.53 \\ 0.48 & 0.47 \end{pmatrix} \quad \begin{pmatrix} 0.52 & 0.53 \\ 0.48 & 0.47 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.52 \\ 0.48 \end{pmatrix}$$

# Quantum Microwave

Quantum Digital Coffee



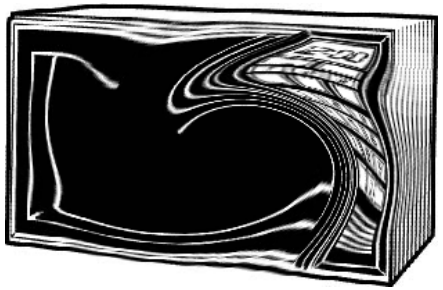
**Scalding Hot**

**H**



**Freezing Cold**

**C**



Quantum Microwave (QM)

What are the rules for the Quantum Microwave?

# The Amplitude Attitude

**H** **C**

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$|0\rangle$ $|1\rangle$

qubit

$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$\alpha|0\rangle + \beta|1\rangle$

Complex numbers $(a + bi)$ ■

$i = \sqrt{-1}$

If we measure the state then

with probability $|\alpha|^2$ we get H

with probability $|\beta|^2$ we get C

$|\alpha|^2 + |\beta|^2 = 1$

## For Our Purposes

$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$\alpha|0\rangle + \beta|1\rangle$

Real numbers ■

$-1 \leq \alpha \leq 1$

$-1 \leq \beta \leq 1$

$\alpha^2 + \beta^2 = 1$

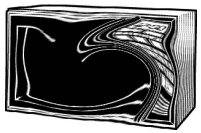
If we measure the state then

with probability $\alpha^2$ we get H

with probability $\beta^2$ we get C

# Unitary

$\neq$ COVENANT OF UNITARIAN UNIVERSALIST PAGANS

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \rightarrow \text{  } \rightarrow \begin{pmatrix} \frac{\alpha+\beta}{\sqrt{2}} \\ \frac{\alpha-\beta}{\sqrt{2}} \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \frac{\alpha+\beta}{\sqrt{2}} \\ \frac{\alpha-\beta}{\sqrt{2}} \end{pmatrix}$$

$$\frac{1}{\sqrt{2}} = 0.70710678...$$

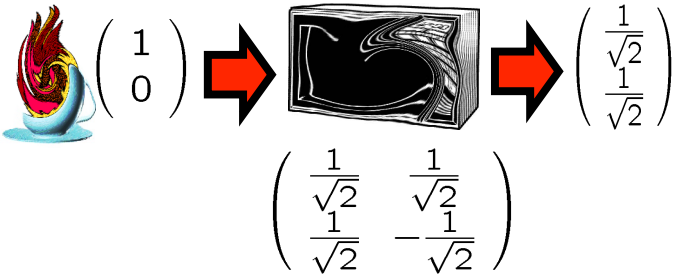
For quantum microwaves, this must be a “Unitary matrix”

Set of conditions (like in Markov where column sums to 1)

Notice some of the entries can now be negative!

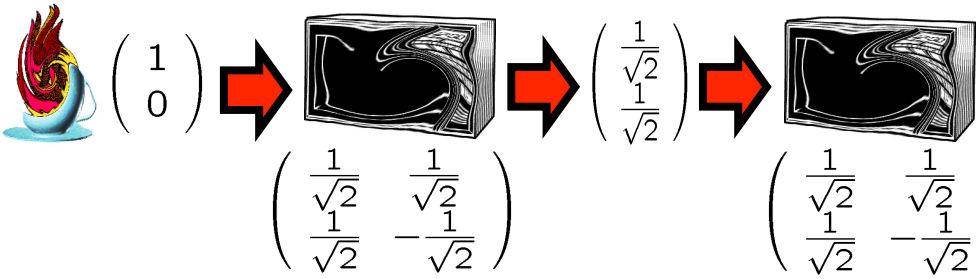
# Interference





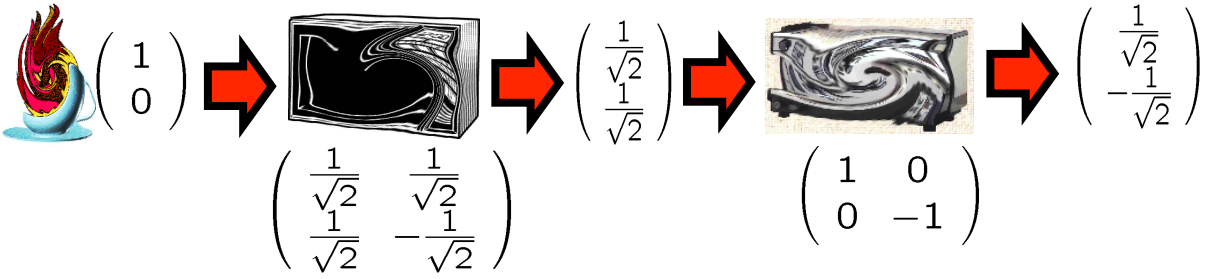
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

50 % **H**
50 % **C**



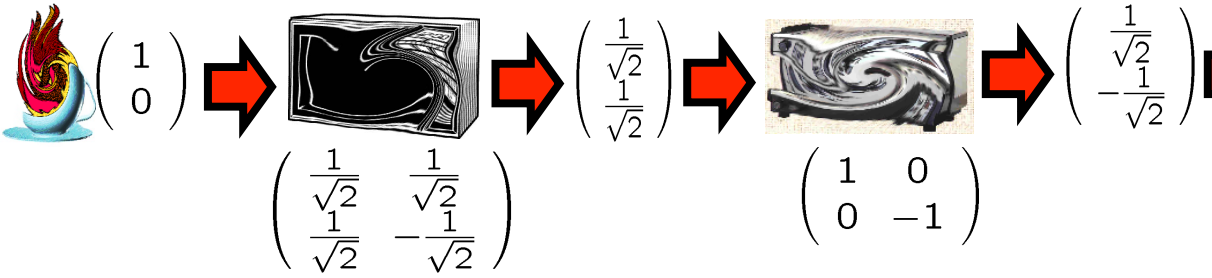
$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

100 %
**H**
0 % **C**



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

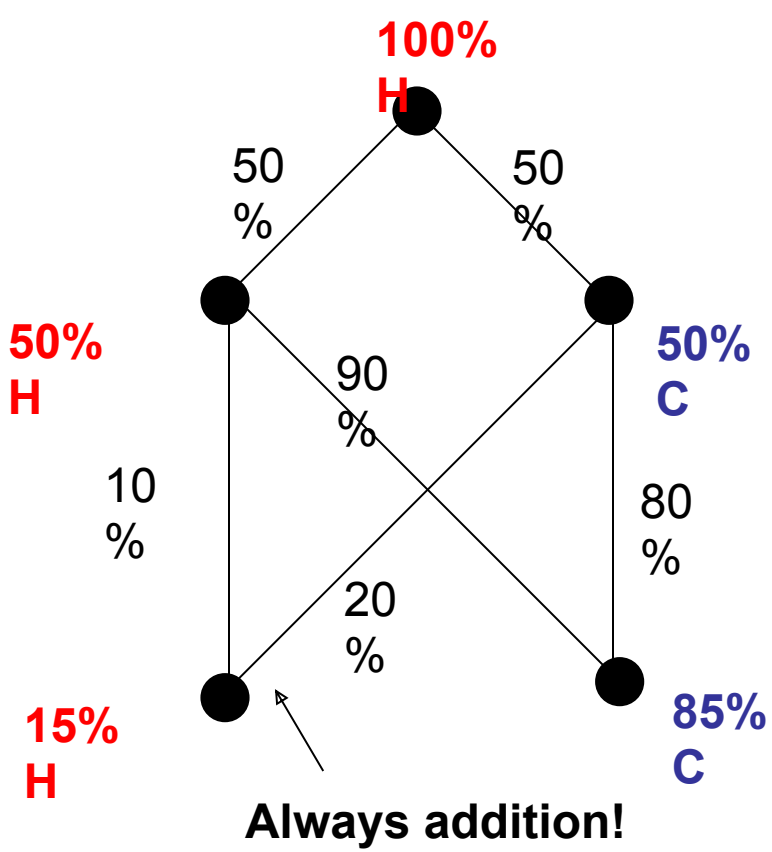
50 % **H**
50 % **C**

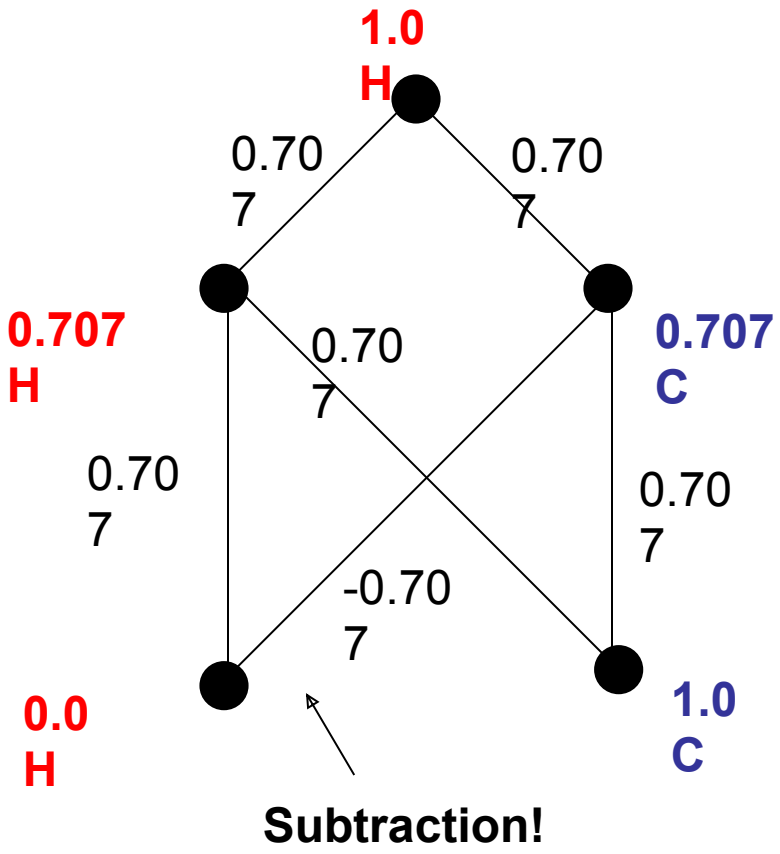


$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

0 % **H**
100 %
**C**

# Interfering Pathways




Classical

Quantum

# Deutsch's Problem



David Deutsch



Dr. Falcon

(1985)

**two qubits**

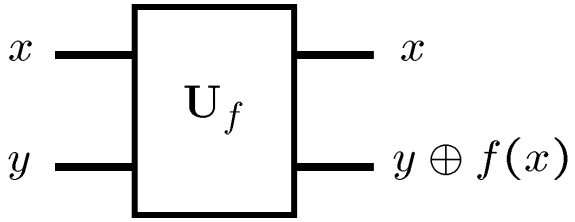
$$\alpha_0|00\rangle + \alpha_1|01\rangle + \alpha_2|10\rangle + \alpha_3|11\rangle$$

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix}$$



Delphi

Oracle




$x, y, f(x) \in \{0, 1\}$

| $x$ | $y$ | $x \oplus y$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

Example $f(x) = x$:

$$U_f = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Four possible functions $f(x)$:

$f(x) = 0$ $f(x) = 1$ $f(x) = x$ $f(x) = \bar{x}$

Constant functions Balanced functions

## Deutsch's Problem

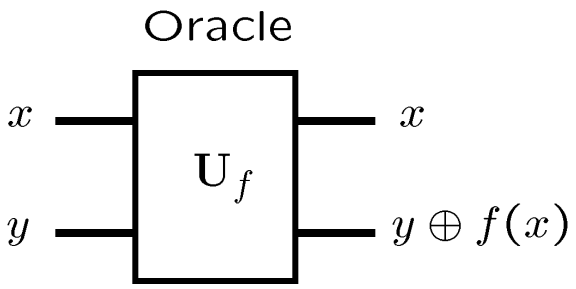
Determine whether f(x) is constant or balanced using as few queries to the oracle as possible.

# Classical Deutsch

Four possible functions $f(x)$:

$f(x) = 0 \quad f(x) = 1 \quad f(x) = x \quad f(x) = \bar{x}$

Constant functions  Balanced functions




$x, y, f(x) \in \{0, 1\}$

Query input of $x_0$ and $y_0$ only gives information about $f(x_0)$.

Knowing $f(x_0)$ not enough to distinguish constant from balanced.

Classically we need to query the oracle <u>two</u> times to solve Deutsch's Problem



Delphi



Delphi

# Quantum Deutsch

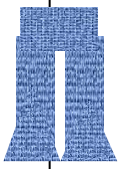


1. Query oracle with $\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$
2. Producing the state $\frac{1}{2}\left(|0f(0)\rangle - |0\bar{f}(0)\rangle + |1f(1)\rangle - |1\bar{f}(1)\rangle\right)$

| $f(x) = 0$ | $f(x) = 1$ | $f(x) = x$ | $f(x) = \bar{x}$ |
|---|---|---|---|
| $\frac{1}{2}(\|00\rangle - \|01\rangle + \|10\rangle - \|11\rangle)$ | $\frac{1}{2}(\|01\rangle - \|00\rangle + \|11\rangle - \|10\rangle)$ | $\frac{1}{2}(\|00\rangle - \|01\rangle + \|11\rangle - \|10\rangle)$ | $\frac{1}{2}(\|01\rangle - \|00\rangle + \|10\rangle - \|11\rangle)$ |
| $\frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}$ | $\frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ -1 \\ 1 \end{pmatrix}$ | $\frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$ | $\frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix}$ |
| Constant functions | | Balanced functions | |

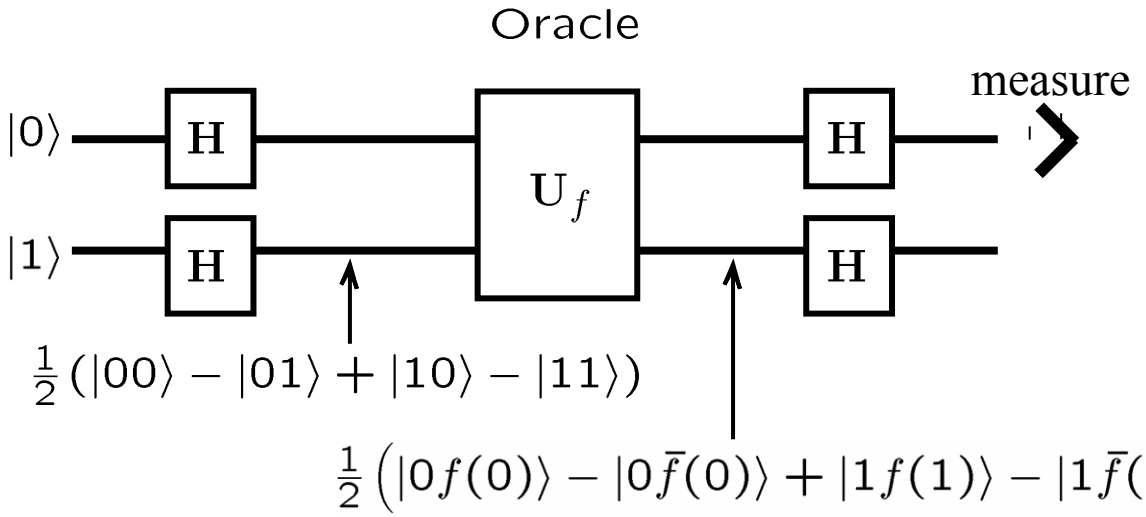
3. Apply the unitary transformation

$$\frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$



| | | | |
|---|---|---|---|
| $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ | $\begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \end{pmatrix}$ | $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$ | $\begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix}$ |
| 100 % $\|01\rangle$ | 100 % $\|01\rangle$ | 100 % $\|11\rangle$ | 100 % $\|11\rangle$ |

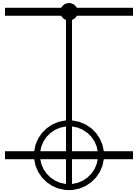
# Deutsch Circuit




$$\mathbf{H} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

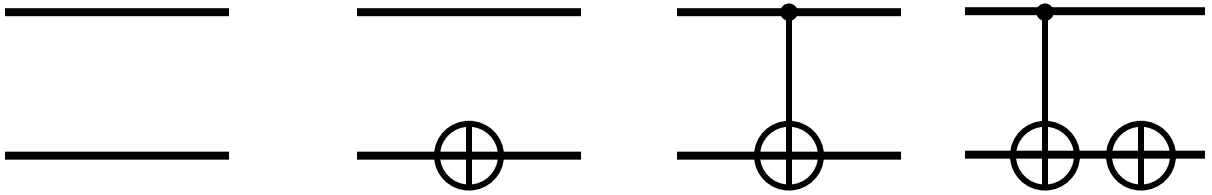


$$= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$



$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$f(x) = 0$ $f(x) = 1$ $f(x) = x$ $f(x) = \bar{x}$



# Deutsch In Perspective



Quantum theory allows us to do in a single query what classically requires two queries!

Interesting but just a factor of 2!

# Mysterious Extra Slide

# Multiplication and Factoring

Multiplication

$$\begin{array}{r} 152 \\ \times\ 421 \\ \hline 152 \\ 304 \\ +\ 1008 \\ \hline 103992 \end{array}$$

multiplying n digit numbers this way
requires about n2 basic operations

Factoring

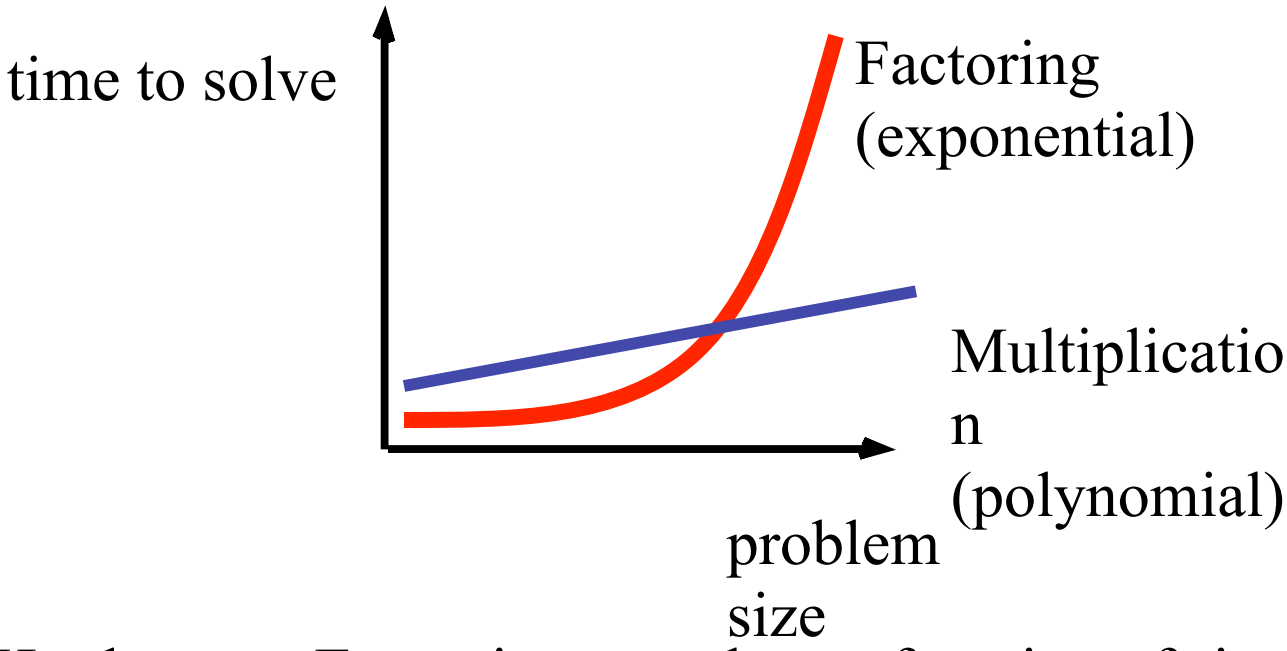
2537

divisible by 2?
divisible by 3?
divisible by 4?
⋮

$2537 = 43 \times 59$

factoring an n digit number this
way requires about 10n/2 basic operations

# Computational Complexity




Hard versus Easy: time to solve as function of size

Does Quantum Computing change Complexity of a Problem?

# Deutsch-Jozsa Problem



(1992)



Given a function $f : \{0,1\}^n \rightarrow \{0,1\}$

$\underbrace{01000111}_{8 \text{ bit}}$

function maps $n$ bit strings to a single bit

$f$ is promised to be either

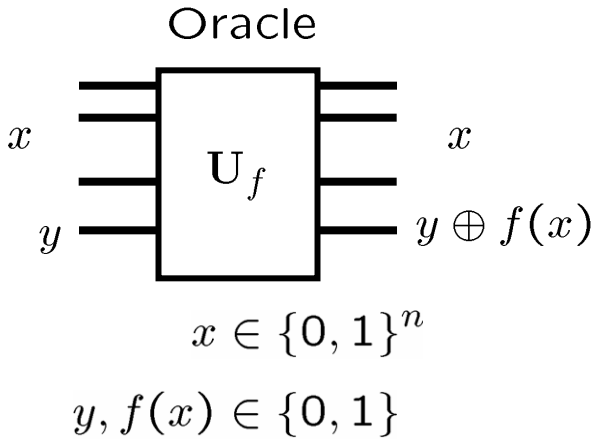
(A) $f$ is constant.

$f(x) = b \ \forall x$

(B) $f$ is balanced.

$f(x) = 1$ if $x \in \mathcal{S}$ otherwise $f(x) = 0$

$\mathcal{S}$ has $2^{n-1}$ elements.




$x \in \{0,1\}^n$

$y, f(x) \in \{0,1\}$

## Deutsch-Jozsa Problem

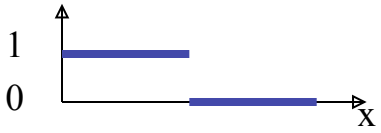
Determine whether f(x) is constant or balanced using as few queries
to the oracle as possible.

# Classical DJ




(A) $f$ is constant.

$$f(x) = b \ \forall x$$



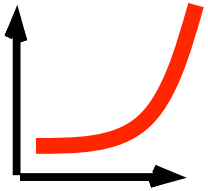

(B) $f$ is balanced.

$f(x) = 1$ if $x \in \mathcal{S}$ otherwise $f(x) = 0$

$\mathcal{S}$ has $2^{n-1}$ elements.

If we never want to be wrong:

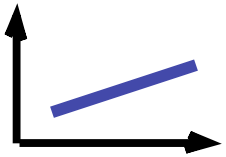
Worst case we need $2^{n-1} + 1$ querries



If we allow a failure probability of $\epsilon$, then

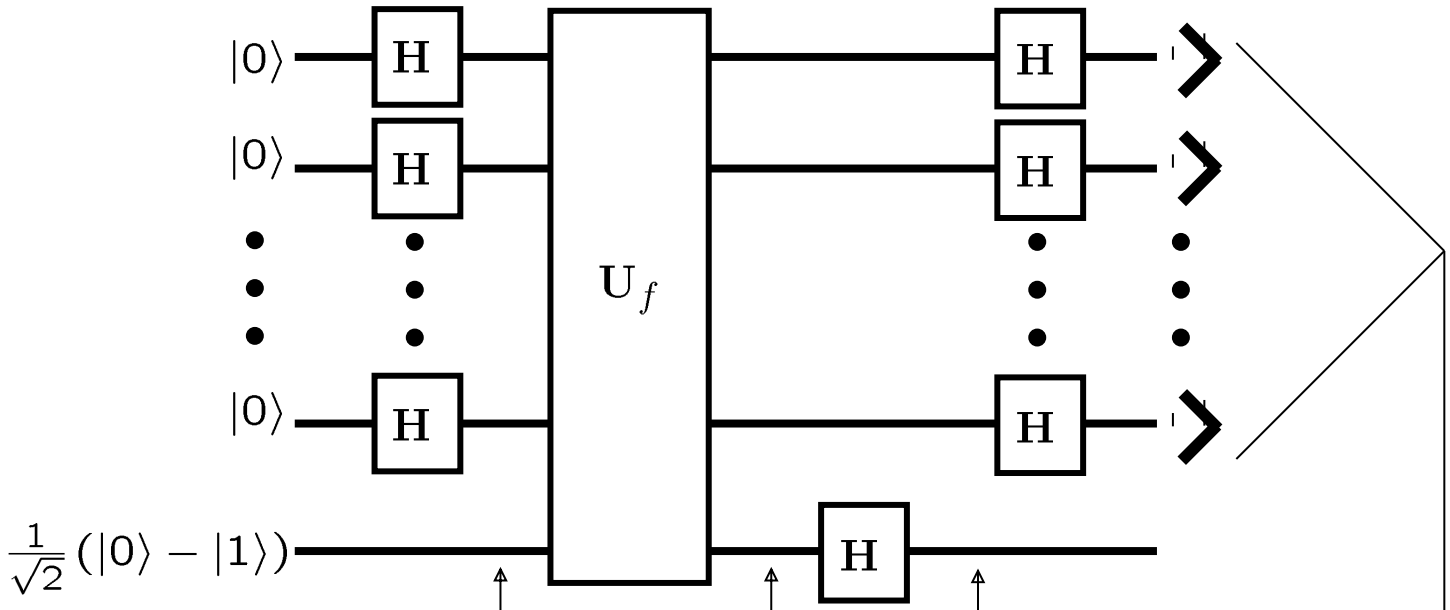
Randomly choose $k$ different $x_k$ to query.

$k = O\left(log\left(\frac{1}{\epsilon}\right)\right)$



Determinstically hard, probabilistically easy.

# Full Quantum DJ




$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|0\rangle - |1\rangle\right)$$

$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|f(x)\rangle - |\bar{f}(x)\rangle\right)$$

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle|1\rangle$$

If all bits are 0, then $f$ is constant.

If all bits are not 0, then $f$ is balanced.

Solves DJ with a SINGLE query vs 2n-1+1 classical deterministic!!!!!!!!

# Simon's Problem

(is that no one does what "Simon says"?) (1994)

Given a function $f : \{0,1\}^n \rightarrow \{0,1\}^n$

function maps $n$ bit strings to a $n$ bit strings

$f$ is promised to satisfy either

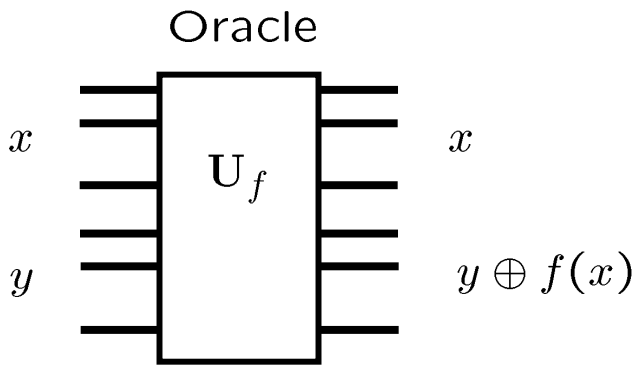
(A) $f$ is distinct on all inputs.

$$f(x) \neq f(y) \quad \forall x \neq y$$

(B) $f$ is distinct up to an XOR mask $s$

$$f(x) = f(x \oplus s)\ \forall x \qquad (s \text{ is unkown and } \neq 0)$$

$$f(x) \neq f(x \oplus t)\ \forall t \neq s$$




## Simon's Problem

Determine whether f(x) has is distinct on an XOR mask or distinct
on all inputs using the fewest queries of the oracle. (Find s)

# Classical Simon

Now both determinsitcally and probabilistically an exponential number of queries is required.

Left as an exercise for those who are asleep.

Make sure to tell the sleepers that they missed a cameo appearance by my good friend Robin Williams.

# Quantum Simon

Requires only number of queries proportional to $n$ the number of digits

# A Brief History of Algorithms

1) Deutsch's algorithm demonstrates task quantum computer can perform in one shot that classically takes two shots.

1992 Deutsch-Jozsa algorithm demonstrates an exponential separation between classical deterministic and quantum algorithms.

1993 Bernstein-Vazirani demonstrates a superpolynomial algorithm separation between probabilistic and quantum algorithms.

1994 Simon's algorithm demonstrates an exponential separation between probabilistic and quantum algorithms.

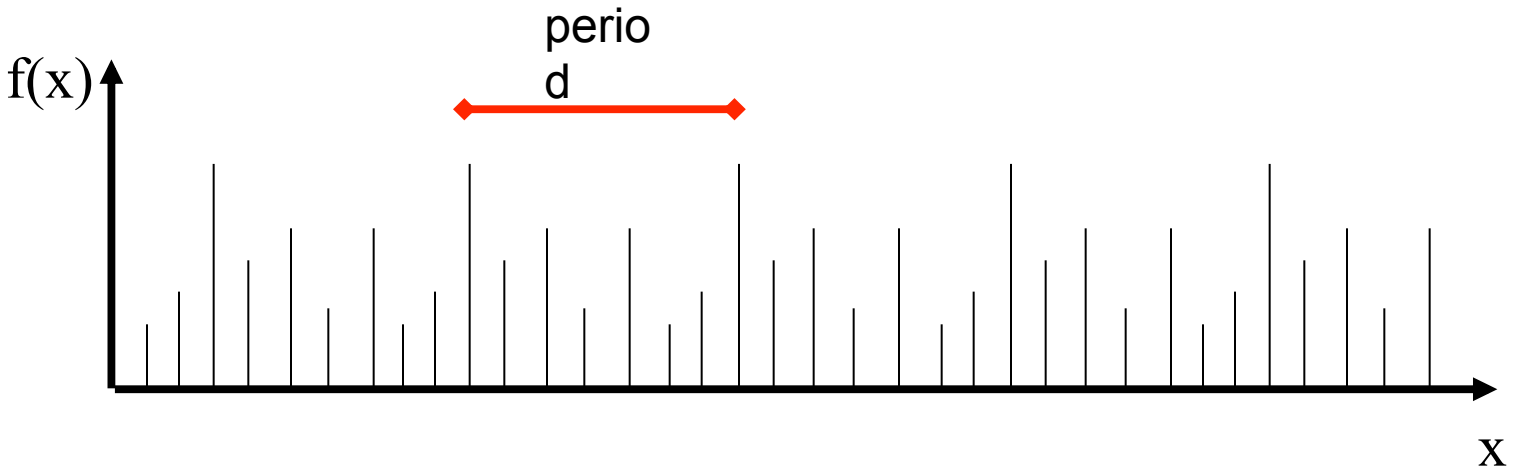
# Quantum Parallelism?

$$\frac{1}{\sqrt{N}}(|\text{Universe 1}\rangle + |\text{Universe 2}\rangle + \cdots + |\text{Universe } N\rangle)$$

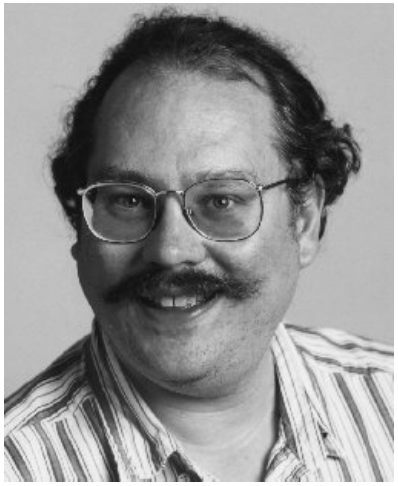
each computes answer, but now...

Many universes collapse when you measure.
You don't get to choose which universe!

Period Finding



# Shor's Algorithm



Peter Shor

188198812920607963838697239461650439807
163563379417382700763356422988859715234
665485319060606504743045317388011303396
716199692321205734031879550656996221305
1687595307650257059

=

398075086424064937397125500550386491199
064362342526708406385189575946388957261
768583317

×

472772146107435302536223071973048224632
914695302097116459852171130520711256363
590397527

Building on the work of Simon, Shor showed how to find the period of a function. This can then be used to factor!

1994

# Shor's Algorithm

Breaks RSA Cryptography!

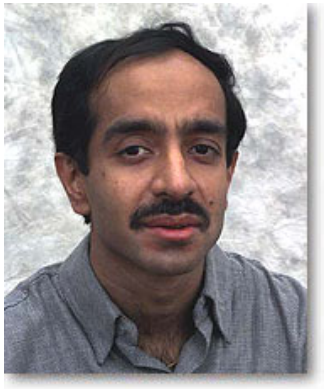


"Only time will tell if and when the problems of building a quantum computer can be overcome….As information becomes the world's most valuable commodity, the economic, political and military fate of nations will depend on the strength of ciphers. Consequently, the development of a fully operational quantum computer would imperil our personal privacy, destroy electronic commerce and demolish the concept of national security. A quantum computer would jeopardize the stability of the world. Whichever country gets there first will have the ability to monitor the communications of its citizens, read the minds of its commercial rivals and eavesdrop on the plans of its enemies. Although it is still in its infancy, quantum computing presents a potential threat to the individual, to international business and to global security. "

-Simon Singh



# Grover's Algorithm



Lov Grover

1996

## Arizona Cacti

*Ferocactus wislizeni*
*Peniocereus greggii*
*Opuntia acanthocarpa*
*Opuntia fulgida*
*Opuntia leptocaulis*
*Opuntia arbuscula*
*Opuntia versicolor*
*Opuntia bigelovii*
*Echinocereus triglochidiatus*
*Echinocereus fasciculatus*
*Mammillaria grahamii*
*Carnegiea gigantea*

Searching a list of n items
for one particular entry

**Classical:** Worst case n queries
**Quantum:** Worst case $\sqrt{n}$ queries

Ex: 100 queries versus 10 queries

# Quantum Communication Complexity



ALICE

**x**

**F(x,y)**

How much (how many bits) do they need to communicate to compute F?



BOB

**y**

Example

**F(x,y)=1 if x=y**

**F(x,y)=0 otherwise**

Alice needs to communicate n bits to Bob

Example

**F(x,y)=x⊕y**

x=010111

y=10010

1 110010 ← **Odd # 1's F=1**

**Even # 1's F=0**

Alice needs to communicate 1 bit to Bob

# Quantum Communication Complexity



ALICE

**x**

How much (how many qubits) do they need to communicate to compute F?



BOB

**y**



Ran Raz (1999)

Exponentially less quantum communication than classical communication!

But currently like Simon 1994, something really useful?

# But What Will It Look Like?

**Solid State**

electron spin in Phosphorus doped Silicon

quantum dots

defects in diamonds

superconducting circuits

**Atomic**

cavity QED

neutral atoms in optical lattices

ion traps

**Molecular**

Liquid NMR

**Photon Based**

linear optics plus single photon devices

# And Why Is It Hard?

# DiVincenzo's Criteria



David DiVincenzo

1. Well defined qubits in a scalable architecture

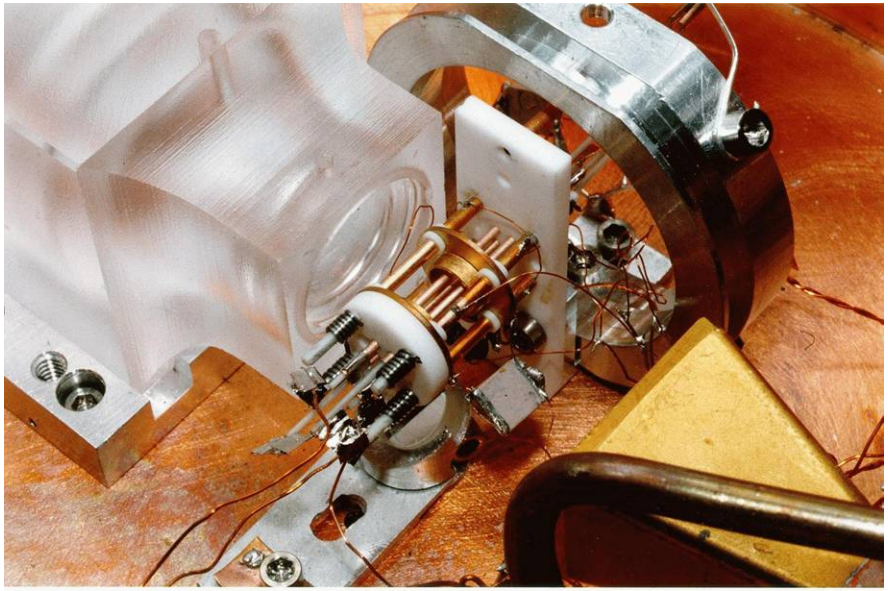
2. The ability to initial the system to a deterministic state.

3. Have faster control over the system than error processes in the system.

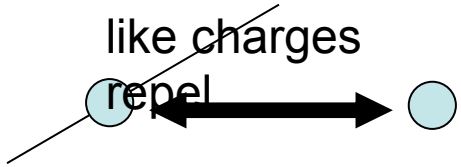
4. Have the ability to perform a universal set of quantum gates.

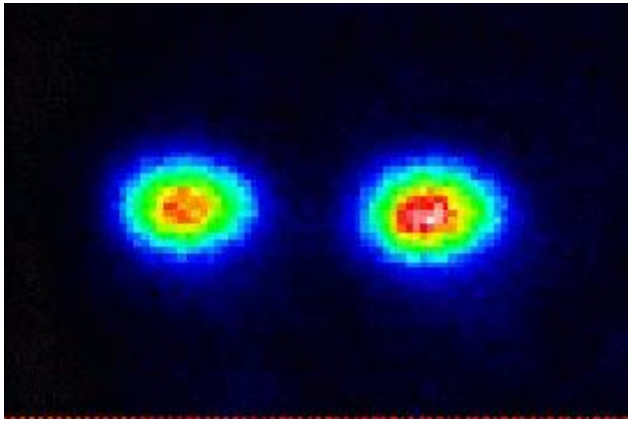
5. Have the ability to perform high quality measurements

# Ion Trap

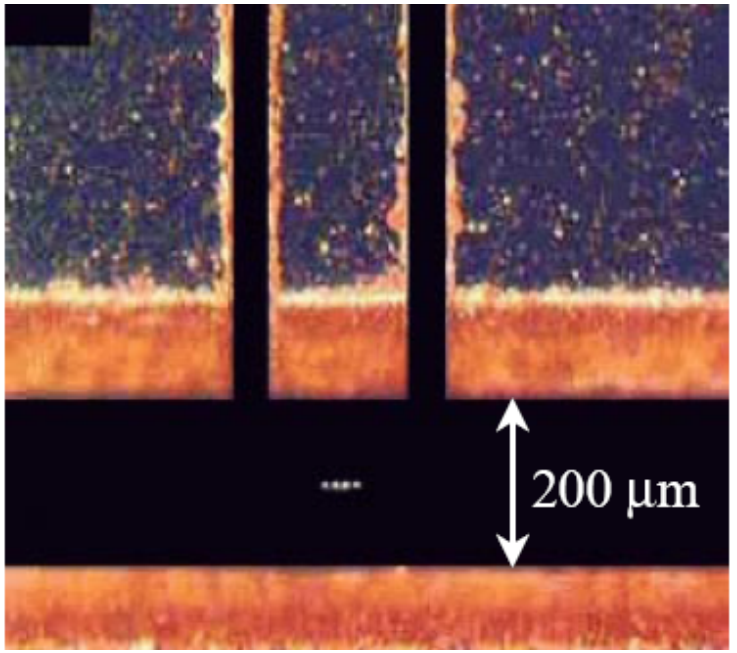


Oscillating electric fields trap ions

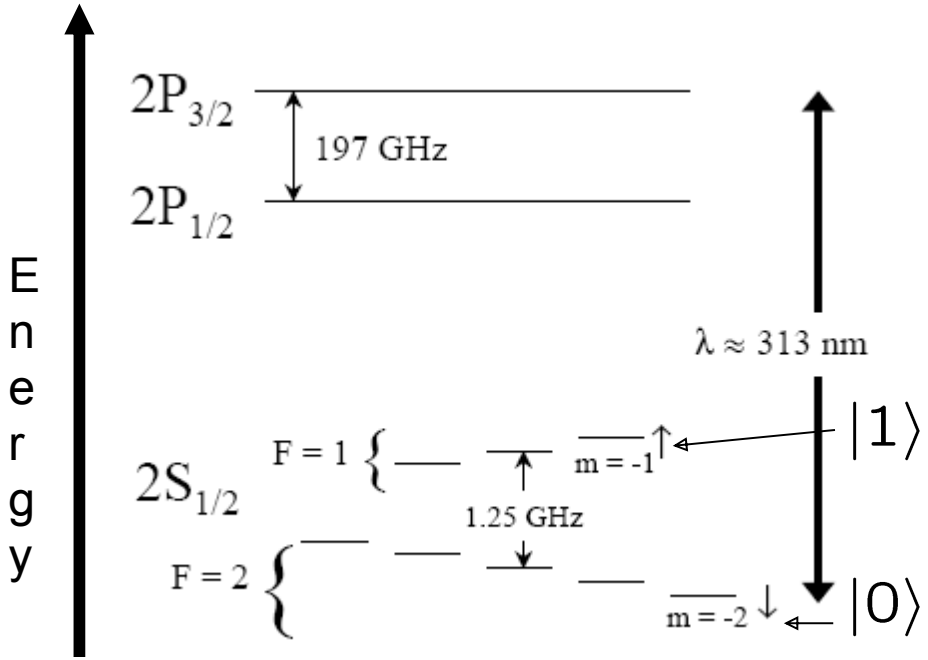


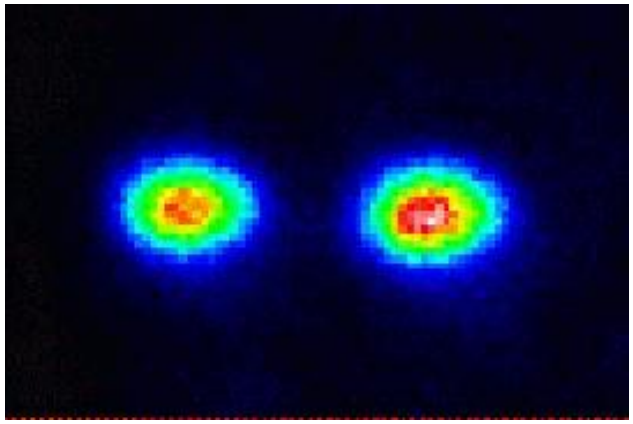



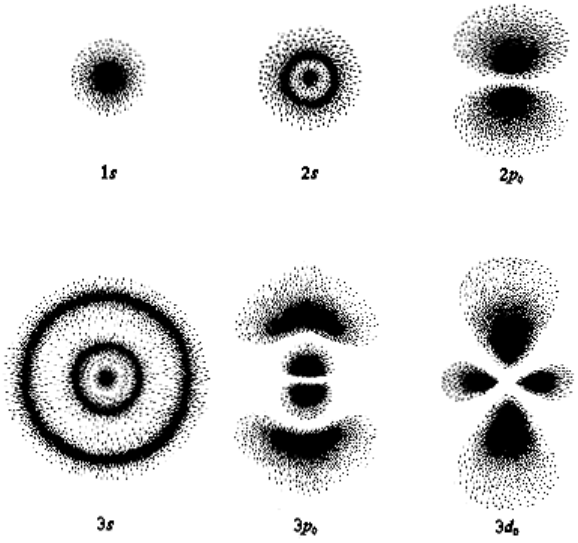
2 9Be+ Ions in an Ion Trap



# Where's the Qubit?



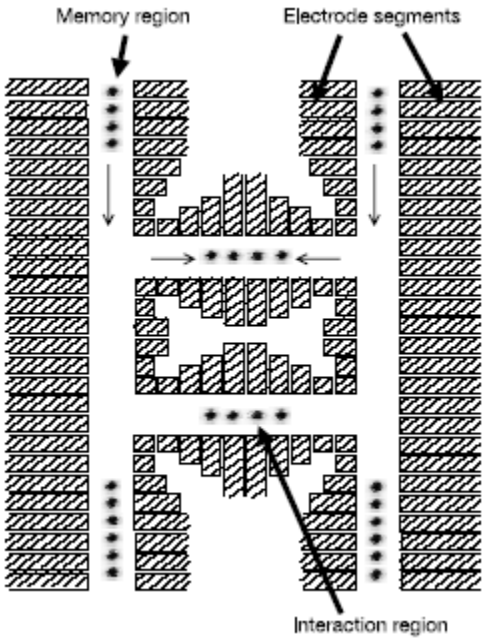



Each ion = 1 qubit

1. Well defined qubits ♠

orbitals

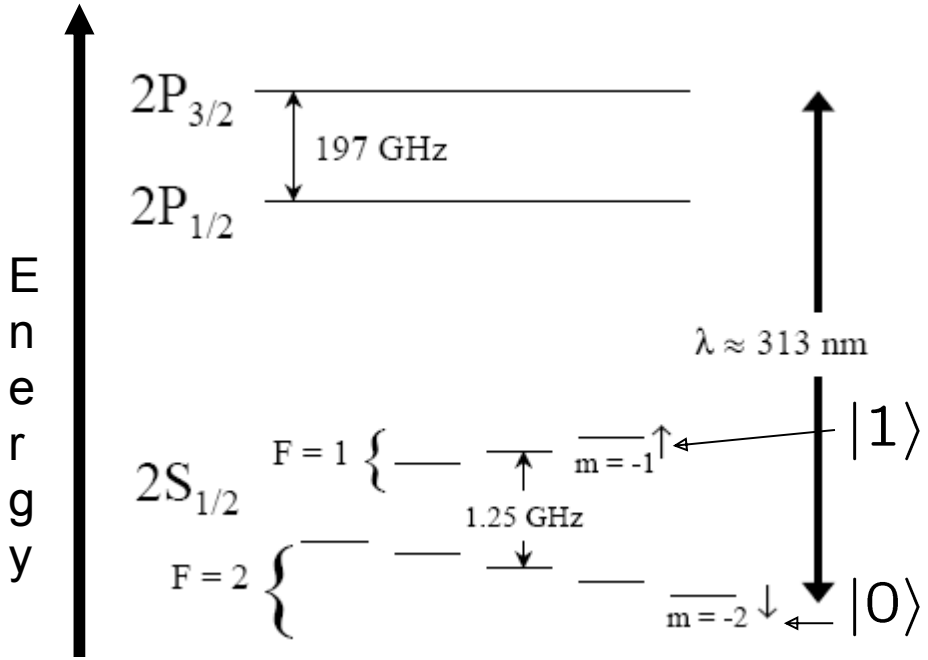
# Scalable?

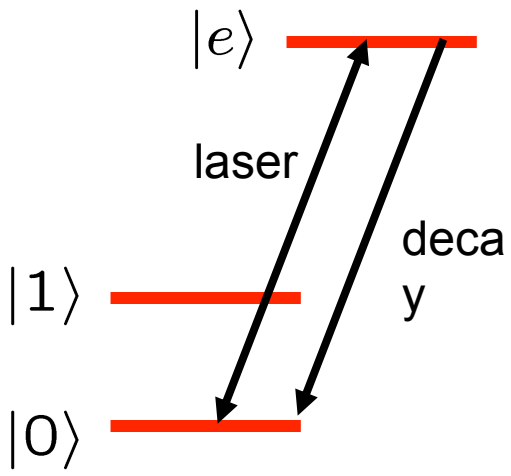


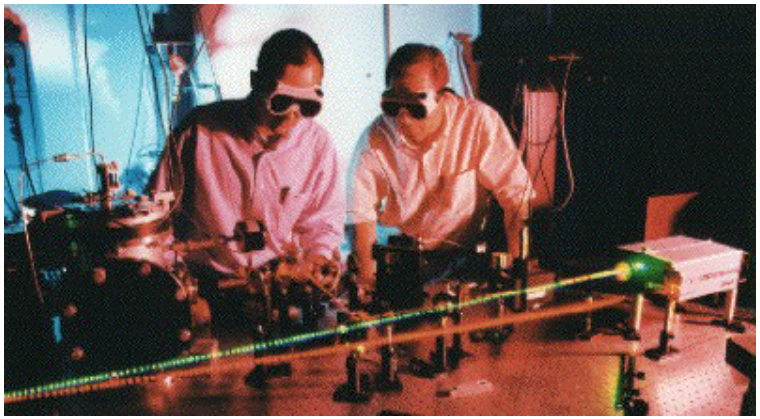

. Well defined qubits in a scalable architecture

Solid state qubits have a huge advantage for scalability.

# Measurement



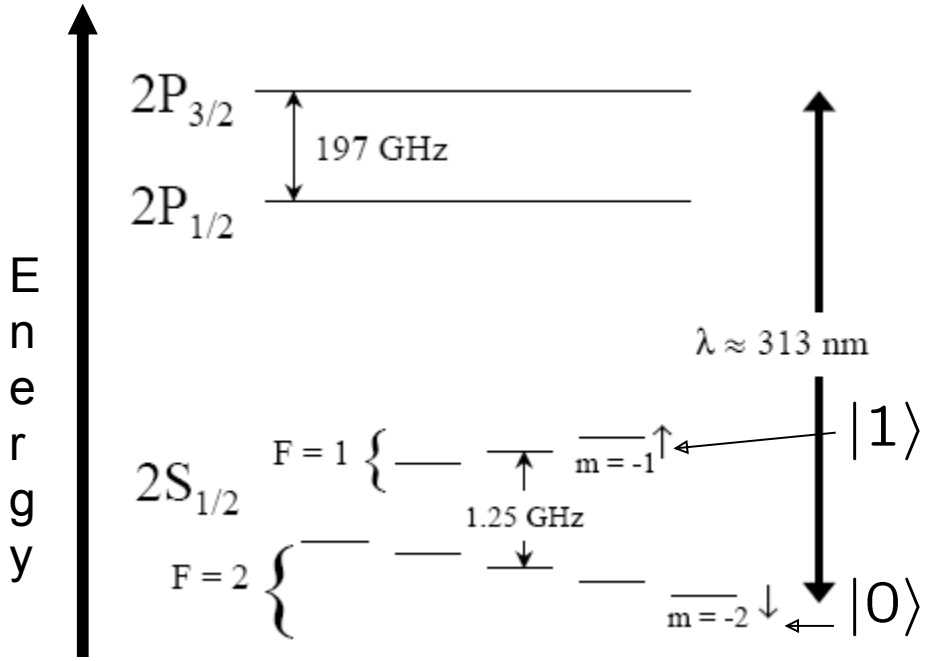



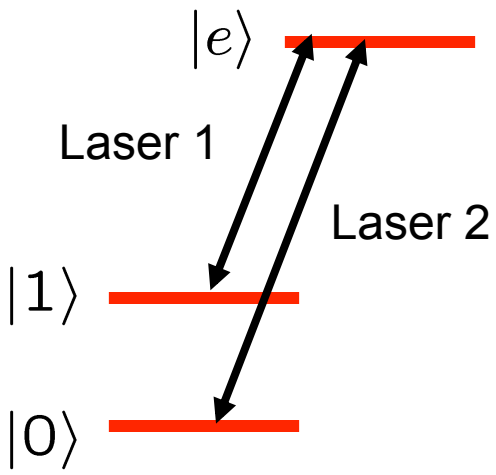
Detecting florescence implies in state 0

99.99% efficiency

5. Have the ability to perform high quality measurements ♠

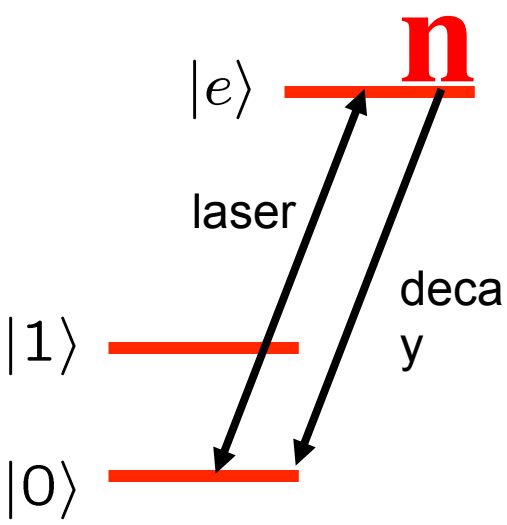
# Single Qubit Operations



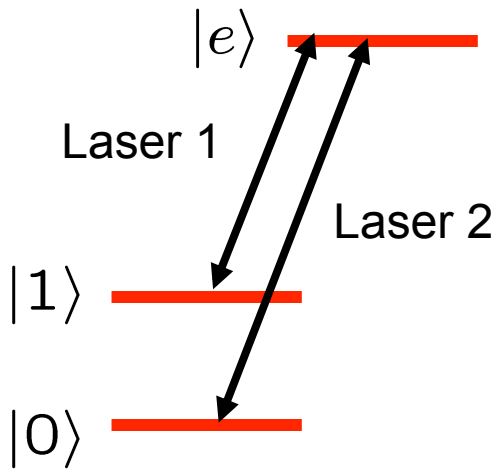




Allows any one qubit unitary operations $\begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}$

# Initialization




measure

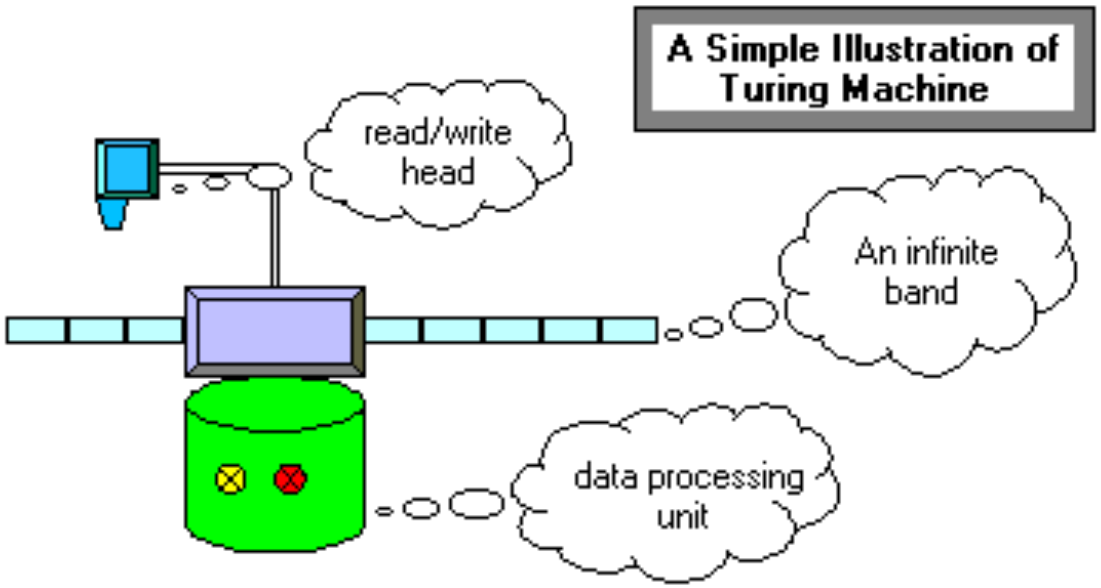



If not in zero state, flip

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

2. The ability to initial the system to a deterministic state. ♠

# Universal C




1. Turing machine reads state of tape at current position.
2. Based on this reading and state of machine, Turing machine writes new symbol at current position and possibly moves left or right.

Certain Turing machines can perform certain tasks.

A Universal Turing Machine can act like any other possible Turing machine (i.e. it is programmable)

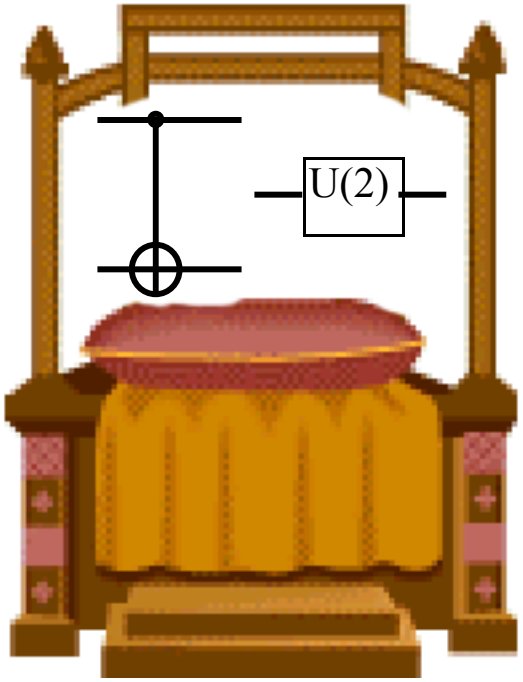
# Universal Quantum Computer

**Universal Quantum Computer**

•a quantum computer which can be programmed to perform any algorithmic manipulation on quantum information.

**Set of Universal Quantum Gates**

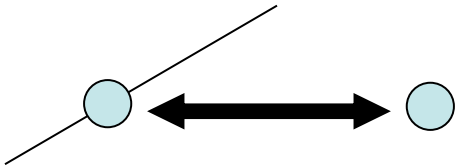
•a set of operations/gates which, acting on the quantum information, can be used to implement (to any desired accuracy) any unitary evolution of the quantum info.



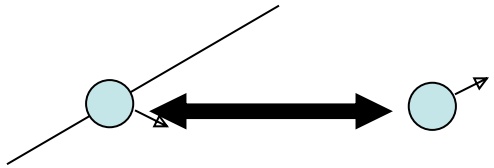

The Royal King and Queen of Universal Quantum Gates

*CNOT and 1-qubit rotations*

# Coupling Two Qubits

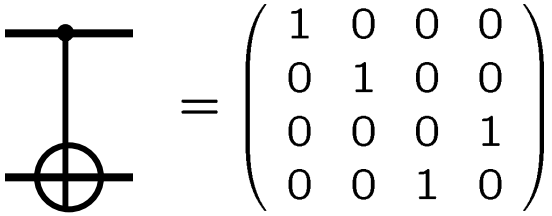


stationary



sloshing mode

These modes can be used as a bus between the qubits.



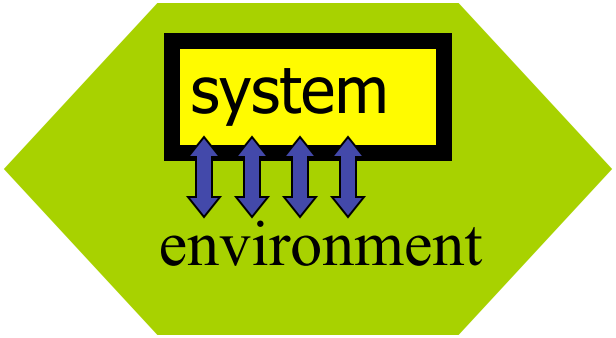
$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

4. Have the ability to perform a universal set of quantum gates ♠

# What is the Problem?

3. Have faster control over the system than error processes in the system.

Real quantum systems are open quantum systems!




Quantum systems readily couple to an environment…

System decoheres:

$|0\rangle$
$|1\rangle$
qubits

$\frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$

**Quantum**

**0**
**1**
bits

50% **0** 50% **1**

**Classical**

**The Decoherence Problem (1996)**

# The Problem

Decoherence is a lot like classical noise, BUT:

Strong coupling to control devices needed to enact computations



Strong coupling to environment causes decoherence

Yingyang of quantum computing

# Quantum Computing is Bunk

Ways Quantum Computers Fail to Quantum Compute

**Decoherence**

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |\text{environment}\rangle \rightarrow \alpha|0\rangle \otimes |\text{environment}_A\rangle + \beta|1\rangle \otimes |\text{environment}_B\rangle$$

**Lack of Unitary Control**

attempting to apply unitary evolution **U** instead results in **V**

or (worse) results in non-unitary evolution

$$\boldsymbol{\rho} \rightarrow \sum_k \mathbf{A}_k \boldsymbol{\rho} \mathbf{A}_k^\dagger$$

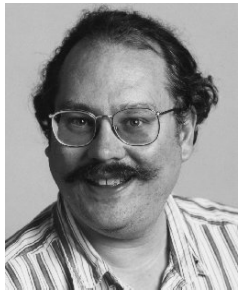
**Measurements are faulty**

measurement result is noisy, incorrect result obtained

**Quantum Computing Disappearing Act**

qubits disappear (leakage of computing states)

# The Quantum Solution (1995-96)



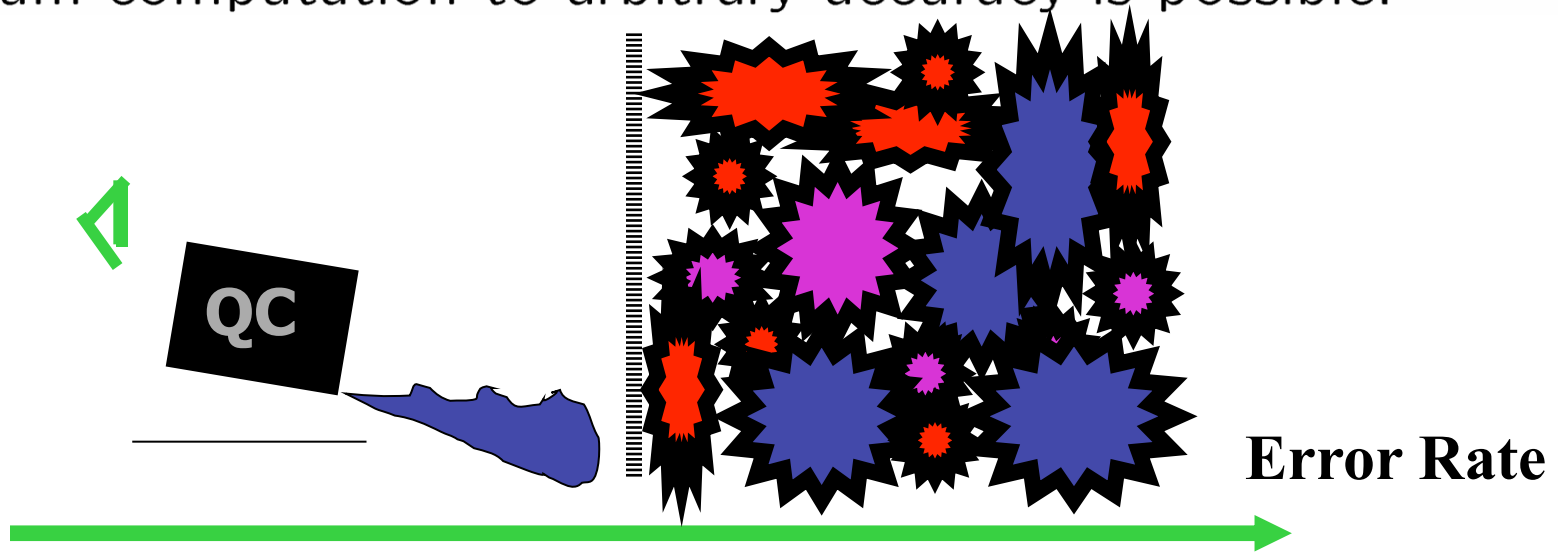


**Fault Tolerant Quantum Computing is Possible!**

Peter Shor  Andrew Steane

**Threshold**

If the error rate (typical number of operations per decoherence or failure event) for single qubit errors is smaller than $\epsilon$, then quantum computation to arbitrary accuracy is possible.

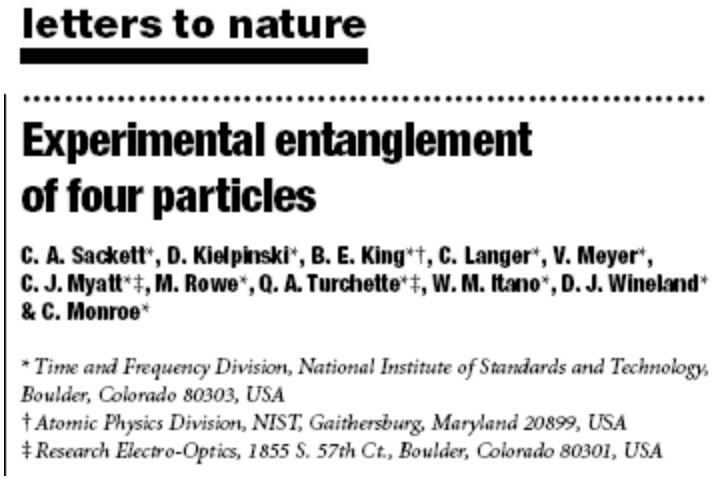


# Ion Trap Parameters

Decoherence rate for qubits: 10 minutes
Gate speed: 10 microseconds
Decoherence rate for bus: 100 microseconds to 100 miliseconds
Measurement errors: 0.01%

3. Have faster control over the system than error processes in the system. ♠

# State of the



letters to nature

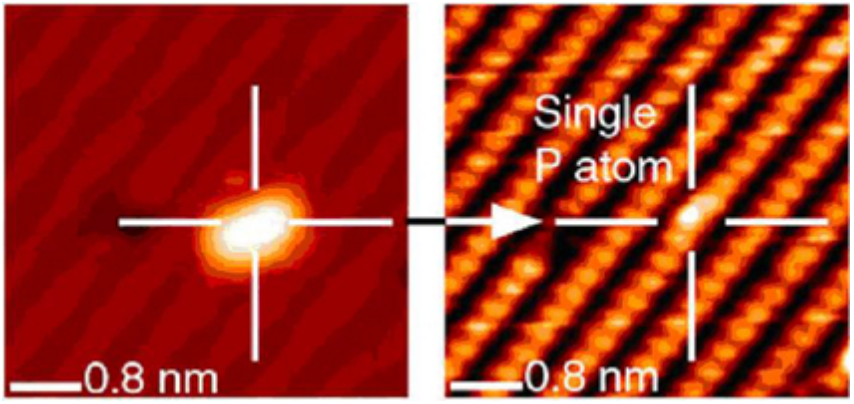
**Experimental entanglement of four particles**

C. A. Sackett*, D. Kielpinski*, B. E. King*†, C. Langer*, V. Meyer*, C. J. Myatt*‡, M. Rowe*, Q. A. Turchette*‡, W. M. Itano*, D. J. Wineland* & C. Monroe*

* *Time and Frequency Division, National Institute of Standards and Technology, Boulder, Colorado 80303, USA*
† *Atomic Physics Division, NIST, Gaithersburg, Maryland 20899, USA*
‡ *Research Electro-Optics, 1855 S. 57th Ct., Boulder, Colorado 80301, USA*

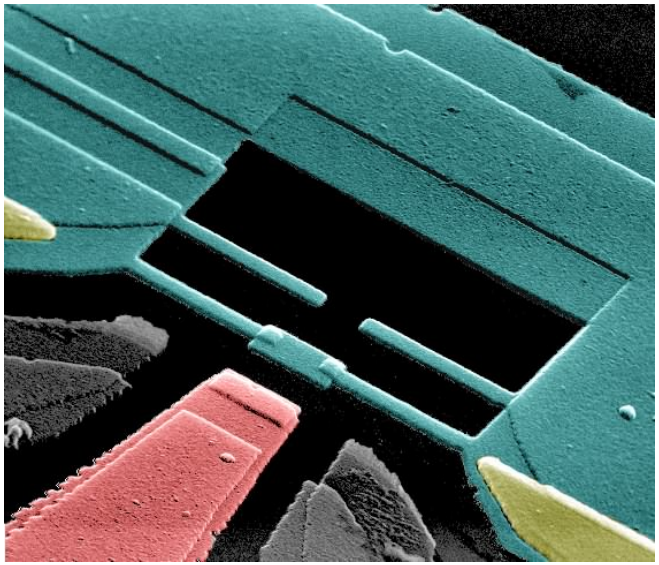
NIST Boulder

# 5 Years, 10 Years




Australia (UNSW)

Electron spin from place P doners

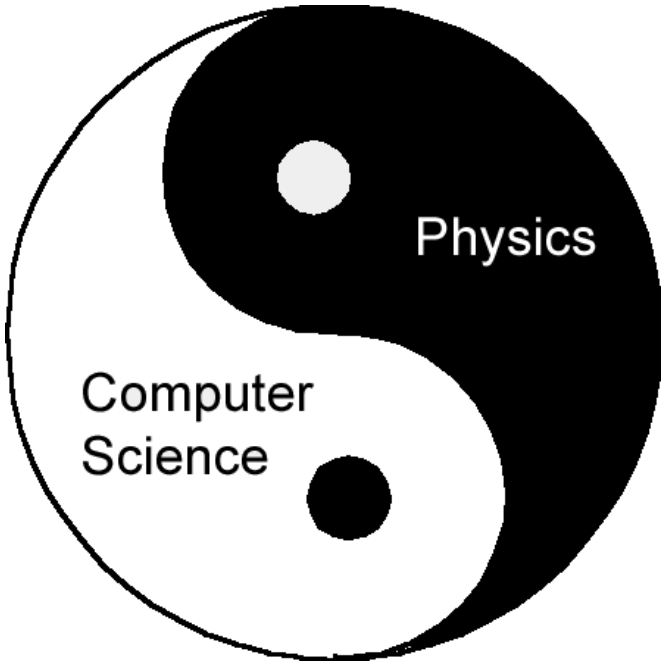


flux qubits in superconducting circuits

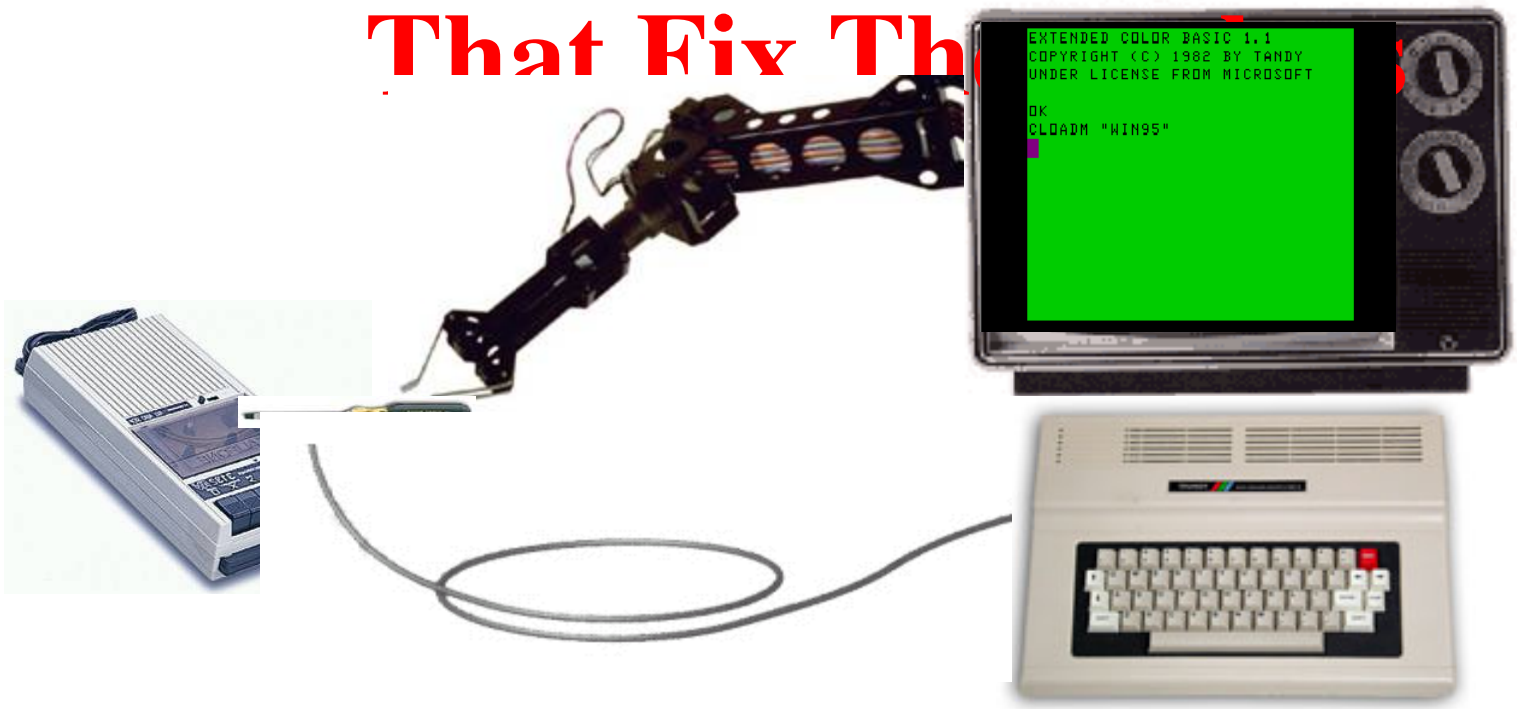
# Fin (Really the Beginning)

When we get to ~30 qubits, a quantum computer can do things today's classical computer cannot do in less than decades of time.



# Quantum Computers That Fix Th



**1941 – First programmable electronic calculator.**
**mechanical relays.**
**1943 – ENIAC.**
**18000 vacuum tubes: “nature abhors the vacuum tube.”**
**1947 – Bell labs develops the transistor.**
**1952 – G. W. Dummer proposes manufacturing electronic equipment in one block with no connecting wires.**
**1959 – Texas Instruments and Fairchild Semiconductor invent the integrated circuit.**
**1964 – first IBM 360 series (the concept of an “architecture”)**
**1964 – Integrated circuit which cost $1000 in 1959 now costs $10. Moore describes his law.**
**1972 – Intel 8008 and 8 bit microprocessor.**
**1975 – First personal computer the Altair.**
**1982 – IBM PC introduced. Personal computer revolution begins.**

# Have We Learned Anything?

Rant mode ON

# The Physics Guarantee

**What is the phase of matter corresponding to the computer?**

PRACTICAL QUESTION (as opposed to philosophical)

**There are distinct PHYSICAL and DYNAMICAL reasons why robust classical computation is possible.**

not all physical systems are equally good for computation:

**there exists systems whose PHYSICS guarantees their ability to enact robust classical computation.**

**Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?**

# Two Paths ("it's made all the difference")

**Hard Drive**



Coding: majority vote of magnetism

Error correction: local energy minimization

Fault-tolerant: macro applied field.

*More than storage?*

**Transistor**

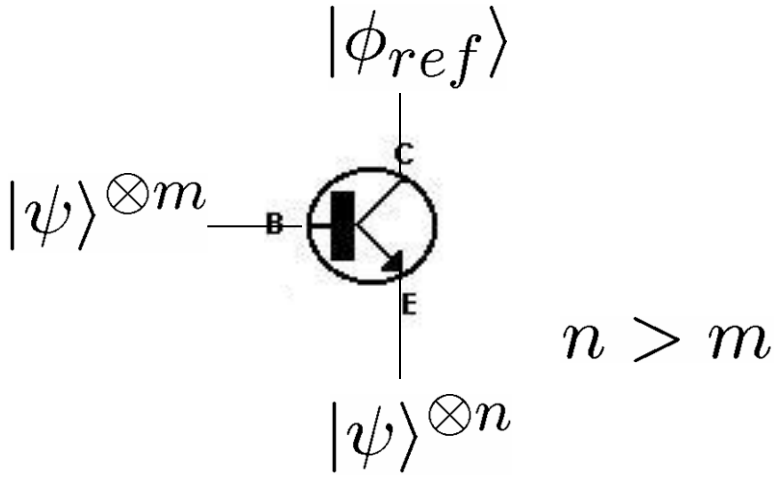


Coding: majority vote of current

Error correction: amplification

Fault-tolerant: conducting / insulating phase transition

# No Quantum Transistor

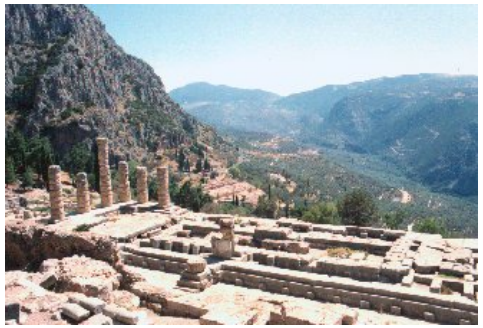






Quantum Cloning Machine

"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

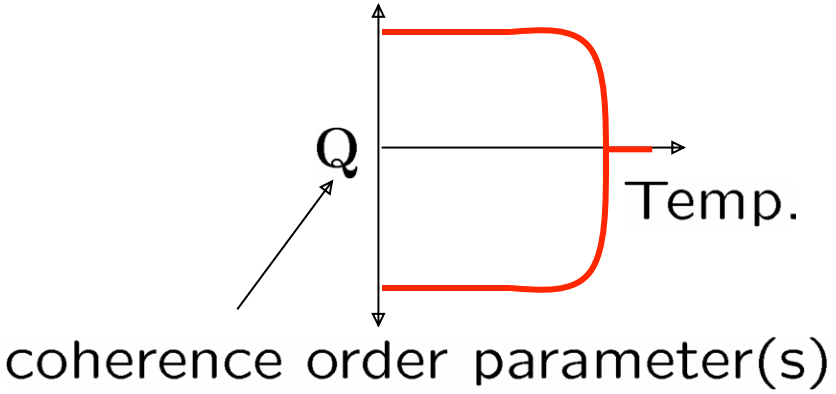


# The Quantum Hard Drive?

(Kitaev)



**Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?**




1. Coherence preserving.

$Q_1 \Rightarrow -Q_1$

macro local operations

2. Accessible Fault-Tolerant Operations

"self-correcting"

$$\left\langle G_1, \ldots, G_{n^k} \right\rangle \cong U(c^n)$$

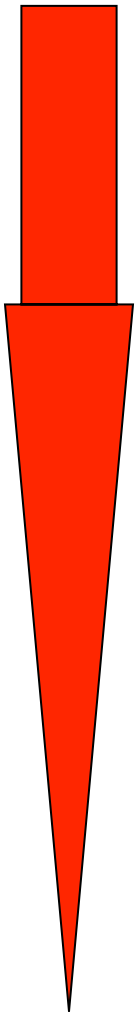
3. Universality

# Fin x2

Error correcting codes
- classical
- quantum

Fault-tolerance
- local structures giving rise to global robustness (CA)

Representation theory
- identification of qubits and logical operations

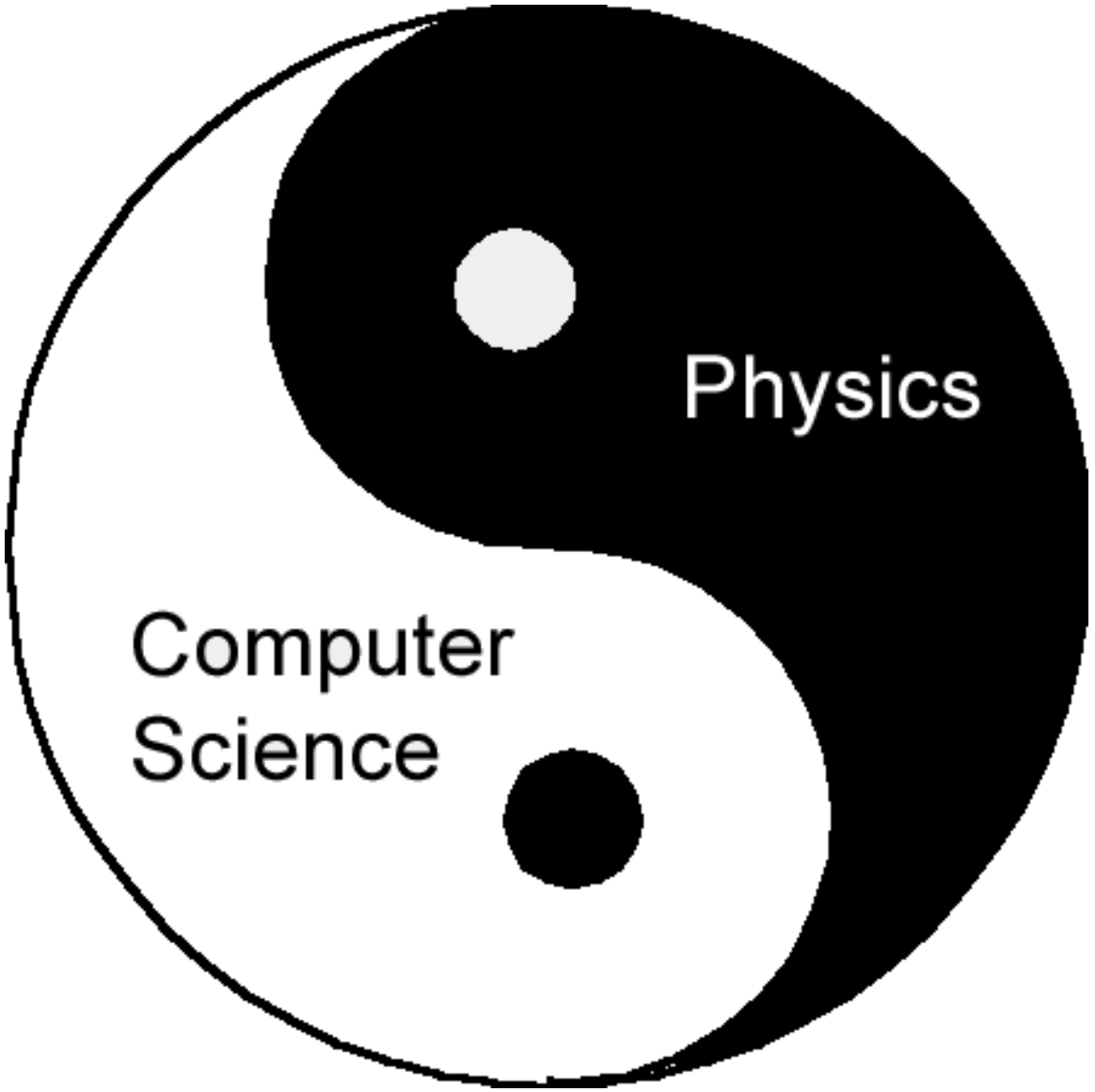


Geometry, Topology?

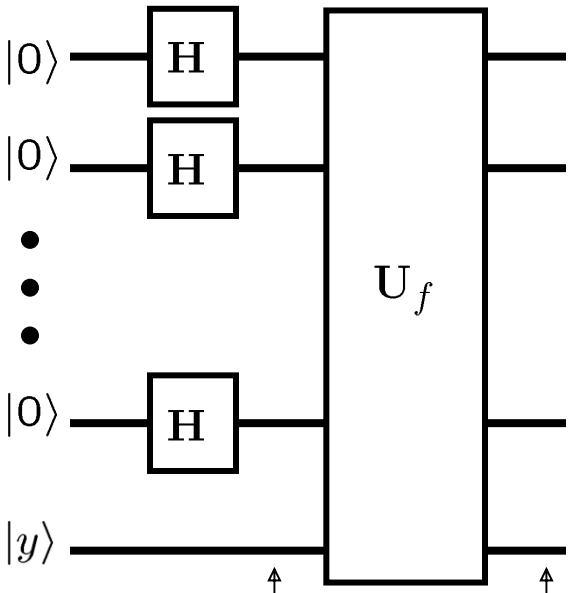
Statistical Physics
- theory of strongly interacting quantum systems
- renormalization group
- methods

Real Physics
- physical implementation
- optical lattices, donor electrons, superconducting circuits, etc.

**Naturally Fault-Tolerant Quantum Computation**

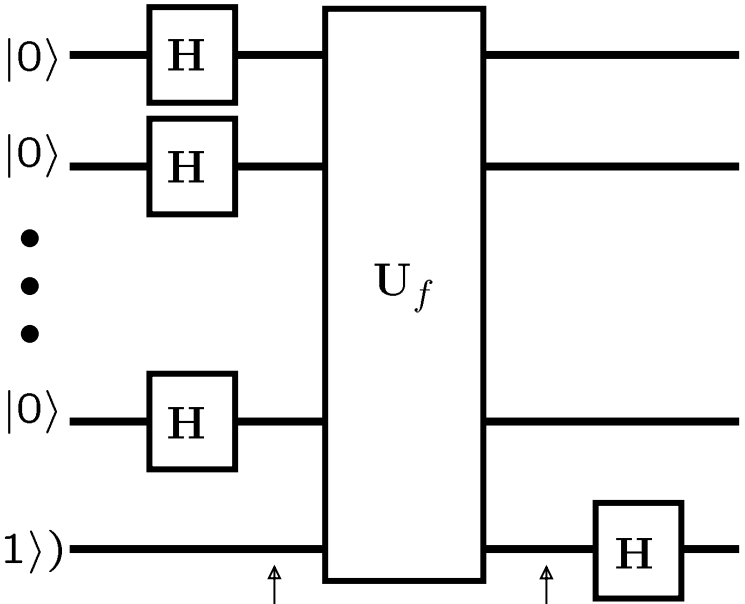

Physics
Computer
Science

# Quantum DJ




$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle|y\rangle \quad \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle|y \bigoplus f(x)\rangle$$

$$|x\rangle = |x_n x_{n-1} \ldots x_1\rangle$$




$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|0\rangle - |1\rangle\right)$$

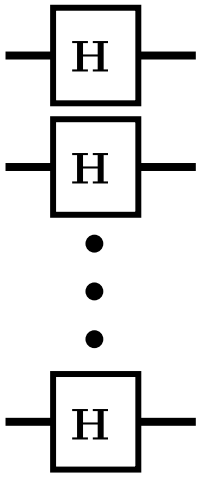
$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x\rangle \left(|f(x)\rangle - |\bar{f}(x)\rangle\right)$$

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle|1\rangle$$

# Quantum DJ

Information about $f(x)$ is in the phase

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle |1\rangle$$




$= \mathbf{H}^{\otimes n}$

$$\mathbf{H}^{\otimes n}|x_n x_{n-1} \dots x_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_n}|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_{n-1}}|1\rangle) \cdots \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_1}|1\rangle)$$

$$= \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x \cdot y} |y\rangle \qquad x \cdot y = x_n y_n + x_{n-1} y_{n-1} + \cdots + x_1 y_1 \mod 2$$

All matrix elements of $\mathbf{H}^{\otimes n}$ are $\pm \frac{1}{\sqrt{2^n}}$

First row of $\mathbf{H}^{\otimes n}$ has all elements $+\frac{1}{\sqrt{2^n}}$

Other rows have equal number $\pm \frac{1}{\sqrt{2^n}}$ elements.

If $f(x)$ is constant, applying $\mathbf{H}^{\otimes n}$ to $\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$ always gives $|0\rangle$.

If $f(x)$ is balanced, applying $\mathbf{H}^{\otimes n}$ to $\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$ always gives a superposition without $|0\rangle$.